higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE
NQF LEVEL 3

12 NOVEMBER 2019

This marking guideline consists of 7 pages.

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QUESTION 1

1.1
- Resistivity
- Temperature
- Cross-sectional area (3)

1.2 $R = \frac{\rho l}{A}$

$l = \frac{RA}{\rho}$

$= \frac{20 \times 0{,}043 \times 10^{-6}}{1{,}2 \times 10^{-6}}$

$= 0{,}717\ m$✓ **OR** 71,7 cm (3)

1.3 The temperature increases with increase in resistance.

OR

Temperature change is directly proportional to change in resistance. (2)

1.4 $R_t = R_0[(1 + \alpha_0(t - \theta)]$

$= 20[(1 + 0{,}00385(65 - 25)]$

$= 23{,}08\ \Omega$✓ (3)

1.5 1.5.1 2
1.5.2 8
1.5.3 4

(3)

1.6 Electrons found in the outermost shell of an atom responsible for bonding/responsible for the chemical properties/responsible for the reaction properties of an atom. (1)

[15]

QUESTION 2

2.1 2.1.1 electromagnetism
2.1.2 relay/contactor/timer
2.1.3 first law of magnetism
2.1.4 right-hand grip

(4 × 1) (4)

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2.2 A motor is an electrical machine which converts electrical energy into mechanical energy.✓ The working of a DC motor is based on the principle that when a current-carrying conductor is placed in a magnetic field, it experiences a mechanical force.✓ The direction is given by Fleming's left-hand rule and hence the conductor moves in the direction of the force.✓

OR

The electromagnet (DC motor) acts like a permanent magnet✓ by producing a magnetic field that always points in the same direction. The commutator/slip ring reverses the coil current so that the coil always spins in the same direction. (3)

2.3 2.3.1 $E = Blv$
$= 0{,}12 \times 0{,}25 \times 3$
$= 0{,}09 \ V$✓ (2)

2.3.2 $F = BIl$
$= 0{,}12 \times 0{,}45 \times 0{,}25$
$= 0{,}0135 \ N$ (2)

2.4 Magnetic flux is a measure of the total✓ magnetic field,✓ while magnetic flux density is a measurement of the strength of a magnetic field✓ at a particular point.✓ (2 + 2) (4)

2.5 2.5.1 Magnetic field
2.5.2 Motion/Force
2.5.3 Induced emf/Current
(3 × 1) (3)

2.6 $H = \frac{NI}{l}$
$= \frac{30 \times 2}{0{,}5}$
$= 120 \ A/m$ (2)
[20]

QUESTION 3

3.1 $E_1 - E_2 = V_1 + V_2 + V_3$

$10 - 4 = 2 + V_2 + 1$

$6 = 3 + V_2$

$V_2 = 3 \ V$ (3)

3.2 3.2.1 $R_p = \frac{R_{L2} \times R_{L3}}{R_{L2} + R_{L3}}$ OR $\frac{1}{R_P} = \frac{1}{R_{L2}} + \frac{1}{R_{L3}}$

$= \frac{5 \times 5}{5+5}$ $= \frac{1}{5} + \frac{1}{5}$

$= \frac{25}{10} = 2{,}5 \ A$ $= \frac{2}{5}$

$R_P = \frac{5}{2} = 2{,}5$

$R_T = R_P + R_{L1}$

$= 2{,}5 + 5$

$= 7{,}5 \ \Omega$

(3)

3.2.2 (a) $R_T = R_{L1} + R_{L3}$

$= 5 + 5$

$= 10 \ \Omega$✓

(2)

(b) $I_T = \frac{V_T}{R_T}$

$= \frac{10}{10}$

$= 1 \ A$

(2)

3.2.3
- The resistance of lamp L1 will remain the same (5 Ω).
- The resistance of lamp L2 will be 0 Ω (also becomes a short circuit).

(2)

3.3 Capacitance is the ability of a capacitor to store an electric charge. (2)

3.4 3.4.1 $C_P = C_1 + C_2$

$= 4 + 2$

$= 6 \ \mu F$

$$C_T = \frac{C_P \times C_3}{C_P + C_3}$$
$$= \frac{6 \times 3}{6+3}$$
$$= 2 \ \mu F$$

∨

$$\frac{1}{C_T} = \frac{1}{C_P} + \frac{1}{C_3}$$
$$= \frac{1}{6} + \frac{1}{3}$$
$$= \frac{3}{6}$$
$$= \frac{6}{3}$$
$$C_T = 2 \ \mu F$$ (3)

3.4.2 $Q_T = C_T V_T$

$= 2 \times 10^{-6} \times 12$✓

$= 24 \ \mu C$ (2)

3.4.3 $V_P = \frac{Q_T}{C_P}$

$= \frac{24}{6}$

$= 4 \ V$

$Q_{4\Omega} = C_1 V_P$

$= 4 \times 10^{-6} \times 4$

$= 16 \ \mu C$ (4)

3.5 3.5.1 Secondary
3.5.2 Primary
3.5.3 Primary

(3 × 1) (3)

3.6 3.6.1 The function of a conservator in a transformer is to take up contraction and expansion of oil without allowing it to come into contact with air
OR
To prevent moisture from entering the transformer. (2)

3.6.2
- Natural air cooling
- Forced air cooling (2)

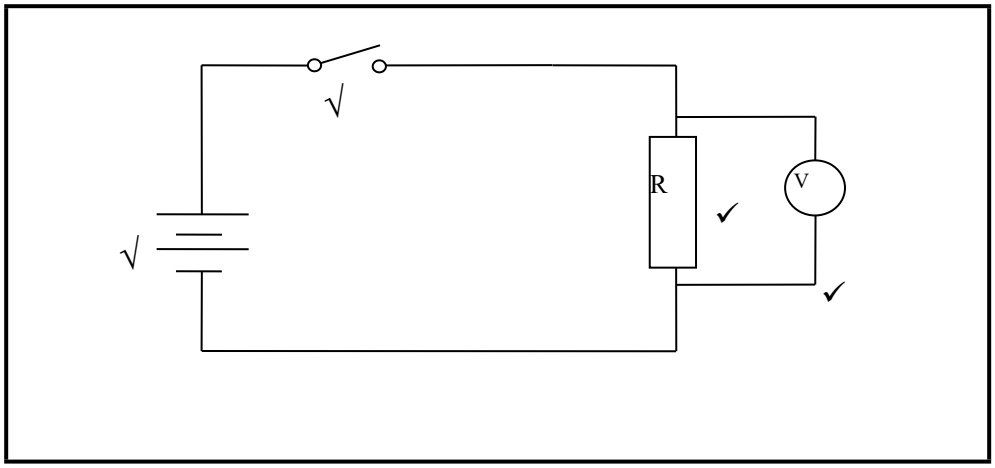
[30]

QUESTION 4

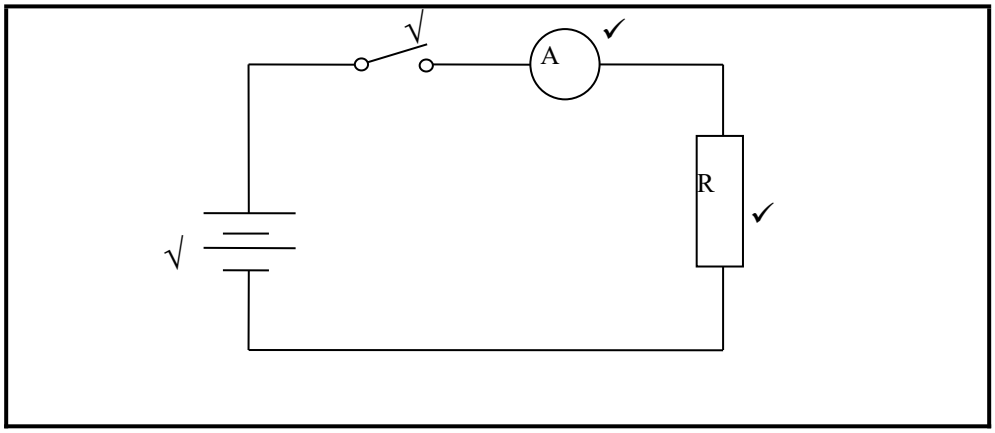
4.1 4.1.1 False
4.1.2 False
4.1.3 True
4.1.4 True
4.1.5 False

(5 × 1) (5)

4.2 4.2.1



4.2.2



(2 × 3) (6)

4.3
- Connect the red test lead to the current (A) socket and the black test lead to the common (com) socket.
- Select amps (A) on the meter selector switch.
- Select the highest reading scale first.
- Connect the ammeter in series with the load to take the measurement. (4)

[15]

QUESTION 5

5.1 Armature reaction is the effect that the magnetic field, produced by the armature current, has on the magnetic field.
OR
It is the effect of the armature ampere turns on the value and the distribution (2)

of the magnetic flux entering and leaving the armature core.

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5.2
- Copper loss
- Iron or core loss
- Friction and winding loss
- Brush contact loss (4)

5.3 5.3.1
- Best power-to-weight ratio of any single-phase AC motor
- High starting torque (Any ONE)

5.3.2
- Poor speed and torque regulation
- Brushes need regular maintenance
- Very noisy (Any ONE)

5.3.3
- Very simple and low cost
- Reliable and quiet
- Speed can be controlled by series resistors (Any ONE)

5.3.4
- Very low efficiency
- Low starting and running torque
- Low power factor (Any ONE)

5.3.5
- It is very quiet.
- Two capacitors optimises efficiency and power factor.
- It was designed for continuous operations.
- Simple construction (Any ONE)

5.3.6
- The value of a run capacitor usually depends on load.
- Low starting torque
- If the start capacitor fails, the motor fails to start. (Any ONE)

(6 × 1) (6)

5.4
- Printing machinery
- Mills
- Cranes
- Hoists
- Pumps
- Lifts
- Generators
- Conveyors (Any relevant 4 × 1) (4)

5.5 $V = E + I_a R_a$

$$I_a = \frac{V - E}{R_a}$$

$$= \frac{220 - 205}{3}$$

$$= 5 \ A$$ (3)

5.6 Phase-failure relay (1)

[20]

TOTAL: 100

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